EDA Assignment — Univariate, Bivariate & Multivariate Analysis

Dataset: Employee Analytics Dataset

Synthetic Dataset creation Link: Activities.ipynb

Dataset Columns

Column	Type	Description
`Department`	Categorical	Employee's department
`Gender`	Categorical	Employee's gender
`Education`	Categorical	Highest education level
`Employment Type`	Categorical	Full Time, Part Time or Contract
`Age`	Numerical	Employee age
`Experience`	Numerical	Years of professional experience
`Salary`	Numerical	Employee salary
`Performance Score`	Numerical	Employee performance score from 1–10
`Projects Completed`	Numerical	Number of projects completed
`Joining Date`	Datetime	Employee joining date

PART A — DATA UNDERSTANDING

Task 1 — Basic Dataset Inspection

1.1 Dataset Dimensions

Find:

- Number of rows
- Number of columns

1.2 Data Types

Display the datatype of every column.

Identify:

- Numerical columns
- Categorical columns
- Datetime column

1.3 Statistical Summary

Use:

```python
df.describe()
```

Also generate an appropriate summary for categorical columns.

1.4 Missing Values

Find the number of missing values in every column.

1.5 Duplicate Records

Find the number of duplicate rows.

PART B — UNIVARIATE ANALYSIS

Task 2 — Department Distribution

Analyze:

```None
Department
```

Requirements:

1. Find the frequency of each department.
2. Calculate the percentage distribution.
3. Create a suitable chart.
4. Identify the most common department.
5. Identify the least common department.

Business Question:

> How is the workforce distributed across departments?

Task 3 — Gender Distribution

Analyze:

```None
Gender
```

Requirements:

1. Find counts.
2. Calculate percentages.
3. Create a suitable visualization.
4. Identify the dominant category.

Business Question:

> What does the employee gender distribution look like?

Task 4 — Education Distribution

Analyze:

```None
Education
```

Requirements:

1. Find counts.
2. Calculate percentages.
3. Create a suitable visualization.
4. Identify the most common education level.
5. Determine whether the distribution is balanced or concentrated.

Task 5 — Employment Type Distribution

Analyze:

```None
Employment Type
```

Requirements:

1. Find counts.
2. Calculate percentages.
3. Create a suitable chart.
4. Identify the dominant employment type.

Business Question:

> What type of employment dominates the organization?

Task 6 — Salary Distribution

Analyze:

```None
Salary
```

Create:

1. Histogram
2. Box plot

Investigate:

- Distribution shape
- Center
- Spread
- Skewness
- Possible outliers

Answer:

1. Where are most salaries concentrated?
2. Is the distribution approximately symmetric or skewed?
3. Are there potential outliers?
4. What does the salary distribution indicate from an HR perspective?

Task 7 — Age Distribution

Analyze:

```None
Age
```

Create:

- Histogram
- Box plot

Answer:

1. What age group is most common?
2. What is the approximate center of the distribution?

3. Is the distribution symmetric or skewed?
4. Are there potential outliers?

Task 8 — Performance Score Distribution

Analyze:

```None
Performance Score
```

Create:

- Histogram
- Box plot

Answer:

1. What range contains most employees?
2. What is the general distribution shape?
3. Are there unusually low or high scores?
4. What does the distribution suggest about overall employee performance?

Task 9 — Projects Completed

Analyze:

```None
Projects Completed
```

Create:

- Histogram
- Box plot

Answer:

1. What number of projects is most common?
2. How widely does project completion vary?

3. Are there potential outliers?

PART C — BIVARIATE ANALYSIS

Task 10 — Experience vs Salary

Analyze:

```None
Experience → Salary
```

Create a scatter plot.

Investigate:

- Direction
- Strength
- Pattern
- Outliers

Answer:

1. Does salary appear to increase with experience?
2. Is the relationship approximately linear?
3. Are there unusual observations?

Business Question:

> Does professional experience appear to be associated with higher compensation?

Task 11 — Pearson Correlation: Experience vs Salary

Calculate Pearson correlation between:

```None
Experience
```

```
Salary
```

Report:

```None
Correlation value:
Direction:
Approximate strength:
Business interpretation:
```

Do not make a causal statement based only on correlation.

Task 12 — Age vs Salary

Analyze:

```None
Age → Salary
```

Create:

- Scatter plot
- Pearson correlation

Compare the result with:

```None
Experience → Salary
```

Answer:

1. Which relationship is stronger?
2. Why might Age and Experience show similar relationships with Salary?
3. Does age itself necessarily explain salary?

Task 13 — Performance vs Projects Completed

Analyze:

```None
Performance Score
Projects Completed
```

Create:

- Scatter plot
- Pearson correlation

Answer:

1. Is there a relationship between performance and project completion?
2. What is the direction of the relationship?
3. How strong is the relationship?
4. Are there any unusual observations?

Task 14 — Joining Year vs Average Salary

Extract the joining year from:

```None
Joining Date
```

Calculate:

```None
Joining Year → Average Salary
```

Create a line plot.

Answer:

1. How does average salary vary across joining years?
2. Is there an increasing or decreasing pattern?
3. Are there any unusual years?

Task 15 — Joining Year vs Average Performance

Create:

```None
Joining Year → Average Performance Score
```

Create a line plot.

Answer:

1. How does average performance vary across joining years?
2. Is there a visible trend?
3. Are there unusual years?

Task 16 — Salary by Department

Analyze:

```None
Department + Salary
```

Create a box plot.

Compare:

- Median salary
- Spread
- Potential outliers

Answer:

1. Which department has the highest median salary?
2. Which has the lowest?
3. Which department has the greatest salary variation?
4. Are there potential outliers?

Task 17 — Salary by Education

Analyze:

```
Education + Salary
```

Create a box plot.

Answer:

1. Does salary appear to differ by education level?
2. Which education group has the highest median salary?
3. Which group has the greatest spread?
4. Are there potential outliers?

Task 18 — Performance by Employment Type

Analyze:

```
Employment Type + Performance Score
```

Create a box plot.

Compare:

- Median
- Spread
- Outliers

Answer:

1. Which employment type has the highest median performance?
2. Which has the lowest?
3. Which group has the greatest variation?

Task 19 — Violin Plot: Salary by Department

Create a violin plot:

```None
Department → Salary
```

Compare the violin plot with the box plot.

Answer:

1. What additional information does the violin plot provide?
2. Can you identify differences in distribution shape?
3. Which department appears to have a wider salary distribution?

PART D — MULTIVARIATE ANALYSIS

Task 20 — Experience + Salary + Department

Create a scatter plot:

```None
X → Experience
Y → Salary
Hue → Department
```

Answer:

1. Is experience positively associated with salary?
2. Do salary levels differ between departments?
3. Does the experience-salary relationship appear similar across departments?
4. Are there unusual observations?

Write one business conclusion involving all three variables.

Task 21 — Experience + Salary + Performance

Create a scatter plot:

```None
X → Experience
Y → Salary
Size → Performance Score
```

Answer:

1. What does point size represent?
2. Do employees with higher performance appear concentrated in particular salary/experience ranges?
3. Are there unusual observations?

Task 22 — Experience + Salary + Education + Gender

Create a scatter plot:

```None
X → Experience
Y → Salary
Hue → Education
Style → Gender
```

Write at least **3 observations**.

Discuss whether the relationship between experience and salary appears different across education or gender groups.

Task 23 — Pair Plot

Create a pair plot using:

```None
Age
Experience
Salary
Performance Score
Projects Completed
```

Investigate:

- Strong relationships
- Weak relationships
- Positive relationships
- Negative relationships
- Possible clusters
- Outliers

Answer:

1. Which pair appears most strongly related?
2. Which pair appears least related?

3. Which variables appear to contain overlapping information?

Task 24 — Pair Plot with Department

Create a pair plot using the same numerical variables with:

```python
hue="Department"
```

Compare it with the previous pair plot.

Answer:

> Does adding Department reveal patterns that were not obvious in the basic pair plot?

Task 25 — Correlation Matrix

Create a correlation matrix for:

```
Age
Experience
Salary
Performance Score
Projects Completed
```

Identify:

1. Strongest positive relationship
2. Weakest relationship
3. Strongest negative relationship, if any
4. Variables that appear to contain overlapping information

Task 26 — Correlation Heatmap

Create a professional correlation heatmap.

Requirements:

- Display correlation values
- Meaningful title
- Clearly labelled axes
- Readable figure size

Answer:

1. Which variables are strongly related?
2. Which variables may contain overlapping information?
3. Which variables appear most strongly associated with Salary?

Task 27 — Department + Education → Average Salary

Calculate:

```
None
Department
+
Education
↓
Average Salary
```

Use:

```python
Python
groupby()
```

Answer:

1. Which department-education combination has the highest average salary?
2. Which has the lowest?
3. Are education-related salary differences consistent across departments?

Task 28 — Department + Gender → Average Salary

Calculate:

```
Department
+
Gender
↓
Average Salary
```

Create a grouped/clustered bar chart.

Answer:

1. How does average salary vary by department?
2. How does the pattern vary across gender groups?
3. Are there departments where the difference between groups appears larger?

Task 29 — Department + Employment Type → Performance

Calculate:

```
Department
+
Employment Type
↓
Average Performance Score
```

Visualize the result using an appropriate chart.

Identify:

1. Highest-performing combination
2. Lowest-performing combination
3. Interesting patterns

4. Any department where employment type appears to make a noticeable difference

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PART E — PIVOT TABLE & HEATMAP

Task 30 — Salary Pivot Table

Create a pivot table:

```
None
Rows    → Department
Columns → Education
Values  → Average Salary
```

Then visualize the pivot table using a heatmap.

Answer:

1. Which combination has the highest average salary?
2. Which has the lowest?
3. Are there departments where education appears to make a larger difference?

Task 31 — Performance Pivot Table

Create a pivot table:

```
None
Rows    → Department
Columns → Employment Type
Values  → Average Performance Score
```

Create:

1. Pivot table
2. Heatmap

Interpret the results.

PART F — FINAL BUSINESS ANALYSIS

Task 32 — Workforce Composition

Using your analysis, answer:

1. How is the workforce distributed across departments?
2. What does the gender distribution look like?
3. Which education level dominates?
4. Which employment type dominates?

Task 33 — Salary Structure

Analyze the salary structure using:

- Salary distribution
- Department
- Education
- Experience

Answer:

1. What does the salary distribution look like?
2. Which departments have higher/lower salary levels?
3. Does education appear to be associated with salary?
4. Does experience appear to be associated with salary?

Task 34 — Experience and Salary

Use:

- Scatter plot
- Pearson correlation
- Multivariate scatter plot

Answer:

> Is experience associated with salary, and does this relationship appear to differ across departments?

Task 35 — Performance and Productivity

Analyze:

- Performance Score
- Projects Completed
- Experience

Answer:

1. Is performance associated with project completion?
2. Is experience associated with project completion?
3. Are employees with greater experience necessarily higher performers?

Task 36 — Department Comparison

Compare departments based on:

- Workforce size
- Salary
- Performance
- Experience
- Education
- Employment Type

Identify at least **5 meaningful differences or patterns**.

Task 37 — Multivariate Business Insights

Identify at least **5 insights involving three or more variables**.

For each insight, use the format:

```None
Observation:
Business Interpretation:
```

```
Supporting Analysis:
```

PART G — FINAL EDA REPORT

Task 38 — Dataset Overview

Include:

- Number of records
- Number of variables
- Numerical variables
- Categorical variables
- Datetime variables
- Missing values
- Duplicate records

Task 39 — Univariate Findings

Provide at least **5 important observations** from univariate analysis.

Task 40 — Bivariate Findings

Provide at least **5 important observations** from bivariate analysis.

Task 41 — Multivariate Findings

Provide at least **5 important observations** from multivariate analysis.

Task 42 — Business Recommendations

Based on the analysis, provide **3–5 recommendations or follow-up questions** for the HR/business team.

Task 43 — Limitations

Mention at least **2 limitations** of the analysis.

Consider:

- Synthetic nature of the dataset
- Sample size
- Missing business variables
- Correlation vs causation
- Possible confounding variables
- Lack of job-level/designation information
- Lack of historical context

FINAL SUBMISSION CHECKLIST

Data Understanding

- Dataset shape
- Data types
- Numerical summary
- Categorical summary
- Missing-value check
- Duplicate check

Univariate Analysis

- Department distribution
- Gender distribution
- Education distribution
- Employment Type distribution
- Salary histogram
- Salary box plot
- Age histogram
- Age box plot
- Performance histogram
- Performance box plot
- Projects histogram
- Projects box plot

Bivariate Analysis

- Experience vs Salary scatter plot
- Experience vs Salary correlation
- Age vs Salary scatter plot
- Age vs Salary correlation
- Performance vs Projects scatter plot
- Performance vs Projects correlation
- Joining Year vs Salary line plot
- Joining Year vs Performance line plot
- Salary by Department box plot
- Salary by Education box plot
- Performance by Employment Type box plot
- Salary by Department violin plot

Multivariate Analysis

- Experience + Salary + Department
- Experience + Salary + Performance
- Experience + Salary + Education + Gender
- Pair plot
- Pair plot with Department
- Correlation matrix
- Correlation heatmap
- Department + Education → Salary
- Department + Gender → Salary
- Department + Employment Type → Performance

Pivot Tables

- Department × Education → Average Salary
- Department × Employment Type → Average Performance
- Salary pivot heatmap
- Performance pivot heatmap

Final Analysis

- 5+ univariate observations
- 5+ bivariate observations
- 5+ multivariate observations